

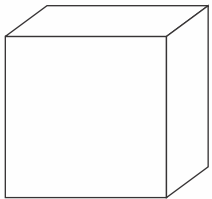
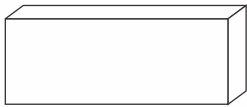
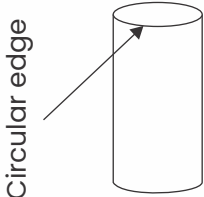
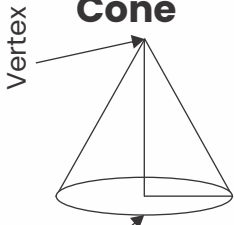
UNIT 12

THREE DIMENSIONAL SOLIDS



THREE DIMENSIONAL FIGURES

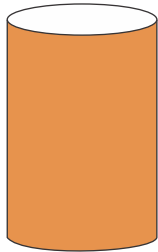
All the objects which have three dimensions length, width, and height are called three dimensional figures or 3D figures.

3D Figure	Property	No. of Faces	No. of Edges	No. of Vertices
Cube 	It is a solid figure whose length, width and height are equal	6	12	8
Cuboid 	It is a solid figure whose length, width and height are not equal	6	12	8
Sphere 	It is a solid figure with complete round surface.	No	No	No
Cylinder 	It is a solid figure with two opposite circular faces and a curved surface.	2 circular faces	2 circular edges	No
Cone 	It is a solid figure with one circular face and a curved surface.	1 circular faces	1 circular edges	1

EXERCISES 12.1

Q1. Name the following 3D shapes and fill in the blanks.

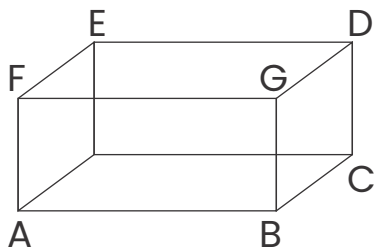
(1)



Name of figure **Cylinder**

Number of circular faces = _____

(2)

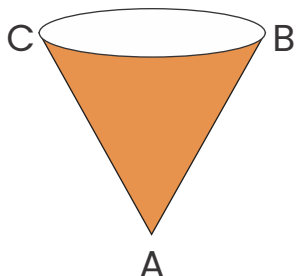


Name of figure **Cuboid**

$\overline{DE}$ represents _____

Points A, B represent _____

(3)

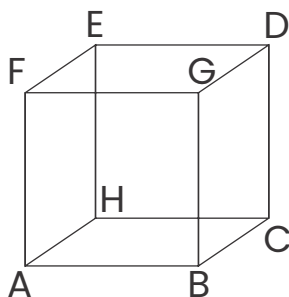


Name of figure **Cone**

Point A represents _____

Q2. Fill in the blanks for the following figures.

(1)



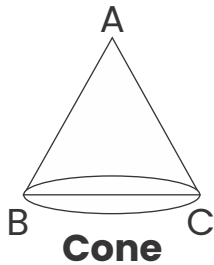
Cuboid

(a) Number of edges = _____

(b) Number of faces = _____

(c) Number of vertices = _____

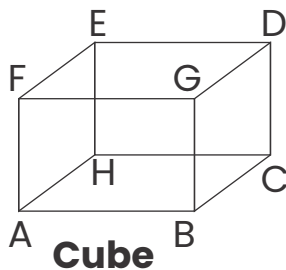
(2)



(a) Point A represents _____

(b) Number of circular faces _____

(3)

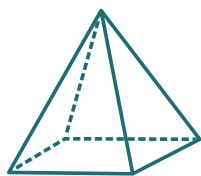


(a) AB represents _____

(b) Point A represents _____

(c) Number of faces _____

Q3. Fill in the name, number of faces, edges, vertices for each shapes.



Name: _____

Faces: _____

Edges: _____

Vertices: _____

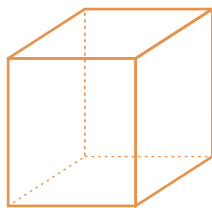


Name: _____

Faces: _____

Edges: _____

Vertices: _____



Name: _____

Faces: _____

Edges: _____

Vertices: _____



Name: _____

Faces: _____

Edges: _____

Vertices: _____

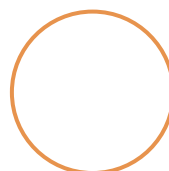


Name: _____

Faces: _____

Edges: _____

Vertices: _____



Name: _____

Faces: _____

Edges: _____

Vertices: _____

SURFACE AREA

The measurement of space occupied by the surface of 3D figures is called surface area.

Units of surface area are sq. cm, sq. m, sq. km etc.

Surface area of cube

We know that length, breadth and height are equal in cube.

So, each surface is of same area.

Surface area of cube = Area of 6 equal faces

$$= 6 \text{ Area of face}$$

$$= 6 \times (L \times L)$$

$$= 6 L^2$$

Example: Find the surface area of cube with each side of 5 cm.

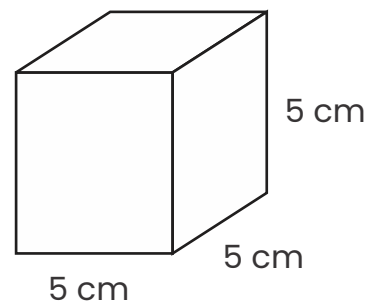
Solution:

Here

Length = $l = 5$ cm

We know that

$$\begin{aligned} \text{Surface Area} &= 6 l^2 \\ &= 6 (5)^2 \\ &= 6 \times 5 \times 5 \\ &= 150 \text{ sq. cm} \end{aligned}$$

**Surface area of cuboid**

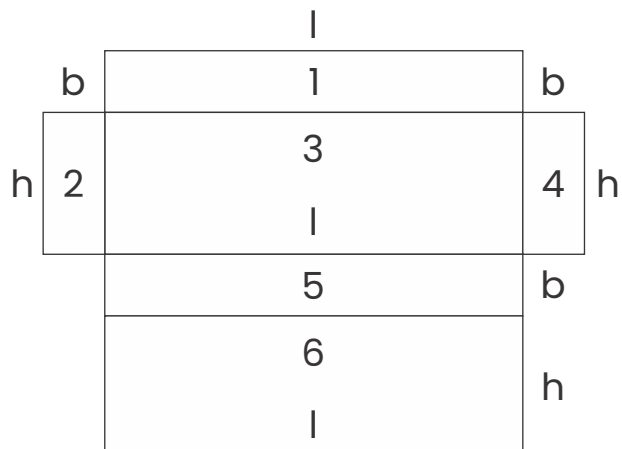
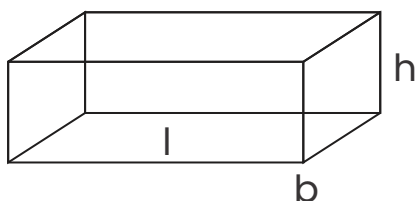
In order to find total surface area of cuboid, we unfold all the six faces as shown in the figure and find the total area of its six faces.

Surface area of faces of cuboid = Area of 1 + Area of 2 + Area of 3 + Area of 4 + Area of 5 + Area of 6

$$= lb + bh + hl + bh + lb + hl$$

$$= 2(lb) + 2(bh) + 2(hl)$$

$$= 2(lb + bh + hl)$$



Example: Find the surface area of a cuboid in which length = 7 cm breadth = 5 cm and height = 6 cm.

Solution:

Here Length = $L = 7$ cm

Breadth = $b = 5$ cm

Height = $h = 6$ cm

$$\begin{aligned}\text{Surface area of cuboid} &= 2(L \times b + b \times h + h \times L) \\ &= 2(7 \times 5 + 5 \times 6 + 6 \times 7) \\ &= 2(35 + 30 + 42) \\ &= 2(107) \\ &= 214 \text{ sq. cm}\end{aligned}$$

VOLUME

Volume is the space occupied by a three dimensional object.

Volume of cube

We know that length, breadth and height are equal in any cube.

So, Volume of cube = $L \times L \times L = L^3$

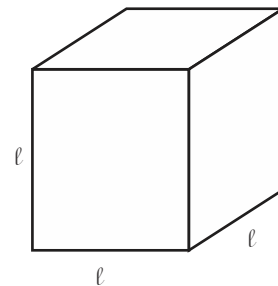
Example: Find volume of a cube whose each side is 5 cm long.

Solution:

Here Length = $l = 5$ cm

We know that

$$\text{Volume of Cube} = l^3 = 5 \times 5 \times 5 = 125 \text{ cm}^3$$



Volume of Cuboid

In order to find formula for volume of cuboid, let us consider a cuboid as shown in the figure.

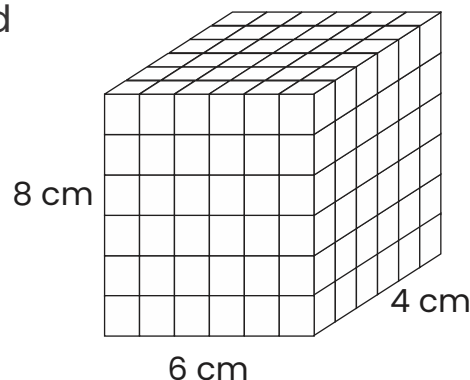
$$\text{Volume of cuboid} = l \times b \times h$$

Example: Find the volume of a cuboid in which length = 8 cm, breadth = 6 cm and height = 4 cm.

Solution:

Here $l = 8$ cm $b = 6$ cm $h = 4$

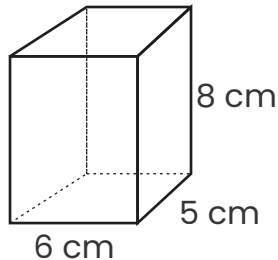
$$\begin{aligned}\text{Volume of cuboid} &= L \times b \times h \\ &= 8 \times 6 \times 4 \\ &= 192 \text{ cubic cm}^3.\end{aligned}$$



EXERCISES 12.2

Q1. Write the units of surface area and volume of the given figures.

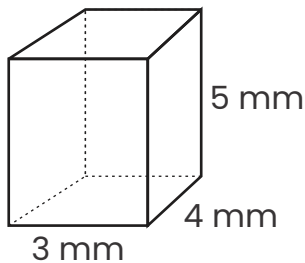
(i)



Unit of surface area = _____

Unit of volume = _____

(ii)

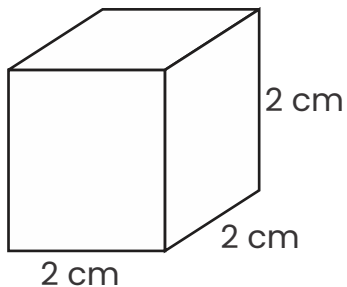


Unit of surface area = _____

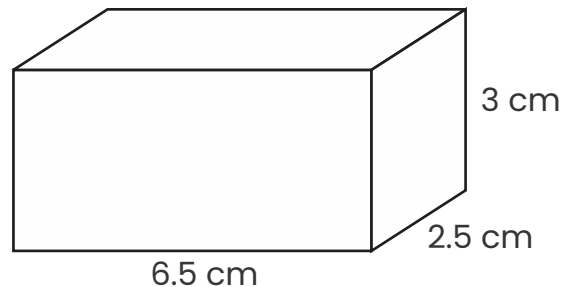
Unit of volume = _____

Q2. Find the surface area and volume of each of the following.

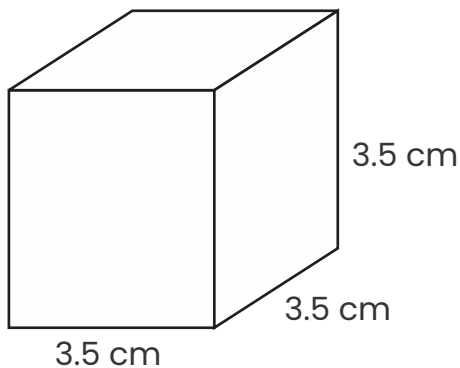
(i)



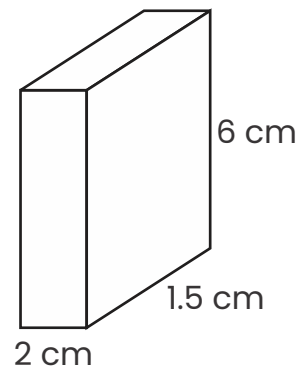
(ii)



(iii)



(iv)



Q3. Find the volume of the following cubes whose each side is:

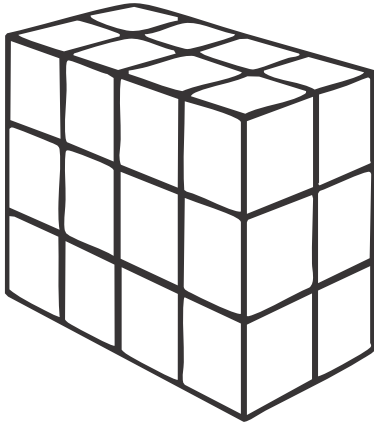
(i) 5cm

(ii) 6.2cm

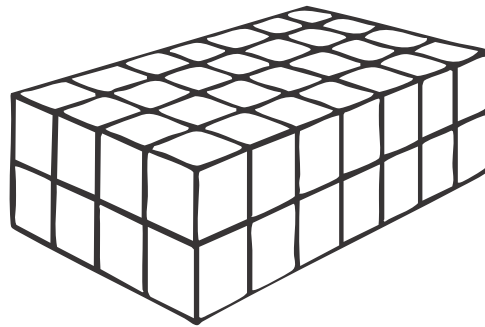
(iii) 2.5cm

Q4. Calculate the volume of the following cuboids.

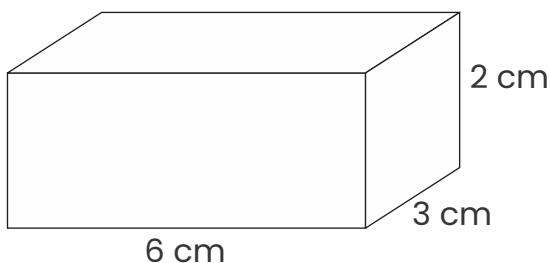
(i)



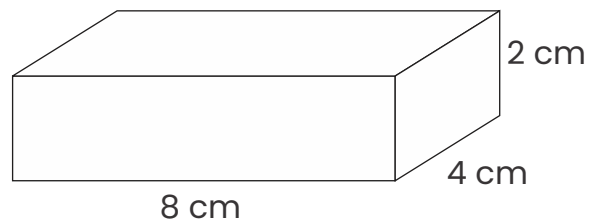
(ii)



(iii)



(iv)



Q5. Find the volume of each of the following cuboids in which:

(i) Length = 9 cm, breadth = 6 cm, height = 4 cm

(ii) Length = 5 cm, breadth = 3.5 cm, height = 4 cm

(iii) Length = 3.5 cm, breadth = 2 cm, height = 2.5 cm

WORD PROBLEMS

- Q1.** Find the cost of polishing cupboard with dimension 3 m, 2 m and 1 m at the rate of Rs 500 per sq. m.
- Q2.** A tank is 5 m long, 4 m wide and 2.5 m high. Find the capacity of the tank?
- Q3.** The dimension of a brick are 25 cm, 10 cm, 4.5 cm. How much space will be occupied by 1000 such bricks?
- Q4.** Find the cost of painting a wooden cubic box with an edge of 2.5 m long at the rate of Rs 50 per sq. m.
- Q5.** Find the capacity of cubical oil tank with each side 5 m long.
- Q6.** How many boxes of dimensions 15 cm, 10 cm, 5 cm are required to fill a box of dimensions 30 cm, 20 cm, 10 cm.